

Lesson 3: 2D motion with rectangular coordinates

1 Position, Velocity, and Acceleration Vectors

Consider a particle that moves in the xy plane as shown below. At some time, t , the particle is located at point P . It's position is specified by the vector $\vec{r}$ connecting the origin O and P .

At some later time, t' , the particle is located at point P' defined by vector $\vec{r}'$. The displacement of the particle from time t to t' is given as

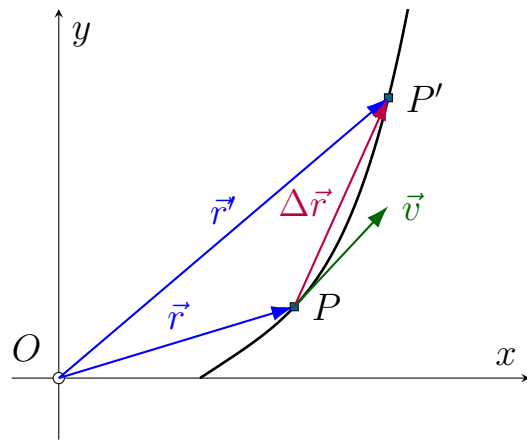
$$\Delta\vec{r} = \vec{r}' - \vec{r}$$

The instantaneous velocity is defined as

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

and is tangent to the path of the particle as shown in the figure below. Similarly the acceleration is given as

$$\vec{a} = \frac{d\vec{v}}{dt}$$



2 Rectangular Components of Velocity and Acceleration

We can express the position vector $\vec{r}$ of a particle in rectangular coordinates as

$$\vec{r} = x(t)\hat{i} + y(t)\hat{j}$$

where $\hat{i}$ and $\hat{j}$ are the unit vectors in the x and y directions, respectively. Differentiating with respect to t we find the following expression for the velocity and acceleration

$$\begin{aligned}\vec{v} &= \frac{d\vec{r}}{dt} = \dot{x}(t)\hat{i} + \dot{y}(t)\hat{j} \\ \vec{a} &= \frac{d\vec{v}}{dt} = \ddot{x}(t)\hat{i} + \ddot{y}(t)\hat{j}\end{aligned}$$

It then follows that the scalar components of velocity and acceleration are

$$\begin{aligned} v_x &= \dot{x}(t) & v_y &= \dot{y}(t) \\ a_x &= \ddot{x}(t) & a_y &= \ddot{y}(t) \end{aligned}$$

Sample Problem 3.1

A drone is flying in the $x - y$ plane. The operator of the drone is located at the origin, O . The motion of the drone is defined by the position vector

$$\vec{r} = (2 + t^3) \hat{i} + (2t^2) \hat{j}$$

where $\vec{r}$ and t are expressed in meters and seconds, respectively. Determine (a) the distance between the drone and its operator at $t = 1$, (b) the distance the drone traveled in the interval from $t = 0$ to $t = 1$ s. (c) the speed and direction of the drone's velocity at $t = 1$ s. (d) the magnitude of the drone's acceleration at $t = 1$ s.

Part (a):

The position of the drone at $t = 1$ is

$$\vec{r}(t = 1) = (2 + 1^3) \hat{i} + (2 \cdot 1^2) \hat{j} = 3 \hat{i} + 2 \hat{j}$$

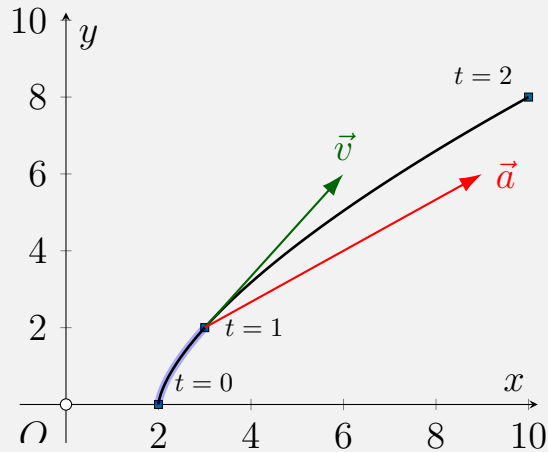
The distance between the origin, O , and the drone is the length of the position vector

$$|\vec{r}(t = 1)| = \sqrt{3^2 + 2^2} = \sqrt{13} = 3.61 \text{ m}$$

Part (b):

This part is the hard part of the problem and could be skipped on a first reading. The figure below shows the trajectory of the drone in the vertical plane. The path shaded in blue is the section traveled between times 0 and 1 seconds.

3.1 – cont



We need to integrate the distance along this curved path. The infinitesimal path length is given as

$$ds = \sqrt{dx^2 + dy^2}$$

where

$$dx = 3t^2 dt \quad dy = 4t dt$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{(3t^2 dt)^2 + (4t dt)^2} = \sqrt{9t^4 + 16t^2} dt$$

Next we integrate along the path.

$$\begin{aligned} \int_0^s ds &= \int_0^1 \sqrt{9t^4 + 16t^2} dt \\ s &= \frac{1}{27} (16 + 9t^2)^{3/2} \Big|_0^1 = \frac{1}{27} (125 - 64) = \frac{61}{27} = 2.26 \text{ m} \end{aligned}$$

3.1 – cont

Part (c):

For this part we need to find the velocity vector.

$$\vec{v} = \frac{d\vec{r}}{dt} = 3t^2\hat{i} + 4t\hat{j}$$

At $t = 1$ we have,

$$\vec{v}(t = 1) = 3\hat{i} + 4\hat{j} = 5 \text{ m/s } \angle 53.1^\circ$$

Part (d):

To find the acceleration we differentiate again,

$$\vec{a} = \frac{d\vec{v}}{dt} = 6t\hat{i} + 4\hat{j}$$

and at $t = 1$ we have,

$$\begin{aligned}\vec{a}(t = 1) &= 6\hat{i} + 4\hat{j} \\ |\vec{a}| &= \sqrt{6^2 + 4^2} = 7.21 \text{ m/s}^2\end{aligned}$$

3 Projectile Motion

For the motion of a projectile the only force acting on the particle is that of gravity. The components of the acceleration are therefore

$$a_x = \ddot{x} = 0 \quad a_y = \ddot{y} = -g$$

We can integrate to find the velocity and position

$$\begin{aligned}v_x(t) = \dot{x}(t) &= v_{x0} & v_y(t) = \dot{y}(t) &= v_{y0} - gt \\x(t) &= x_0 + v_{x0}t & y(t) &= y_0 + v_{y0}t - \frac{1}{2}gt^2\end{aligned}$$

Sample Problem 3.2

An archer fires an arrow at a target that is 100 yards away. The target center is 51.2 inches off the ground. The arrow is released at a height of 70 inches from the ground with a speed of 400 fps. What firing angle α is required to hit the target?

Horizontal motion:

$$v_{x0} = 400 \cos \alpha$$

$$x(t) = x_0 + v_{x0}t = 0 + (400 \cos \alpha)t$$

The time, let's call it t_{hit} when the arrow reaches 100 yd = 300 ft is

$$300 = (400 \cos \alpha)t_{hit} \implies t_{hit} = \frac{300}{400 \cos \alpha}$$
$$t_{hit} = \frac{3}{4 \cos \alpha}$$

Vertical motion:

$$v_{y0} = 400 \sin \alpha$$

$$y(t) = y_0 + v_{y0}t - \frac{1}{2}gt^2 = \frac{70}{12} + (400 \sin \alpha)t - \frac{1}{2}(32.2)t^2$$

3.2 – cont

Projectile hits target:

We want the vertical distance to be 51.1 inches at the time when the arrow hits the target. We therefore set $t = t_{hit}$ and $y(t_{hit}) = 51.2$ inches.

$$\frac{51.2}{12} = \frac{70}{12} + (400 \sin \alpha) \left(\frac{3}{4 \cos \alpha} \right) - \frac{1}{2}(32.2) \left(\frac{3}{4 \cos \alpha} \right)^2$$

Using the identity $1/\cos^2 \alpha = 1 + \tan^2 \alpha$

$$\frac{51.2}{12} = \frac{70}{12} + 300 \tan \alpha - \frac{9}{32}(32.2)(1 + \tan^2 \alpha)$$

Some more rearranging

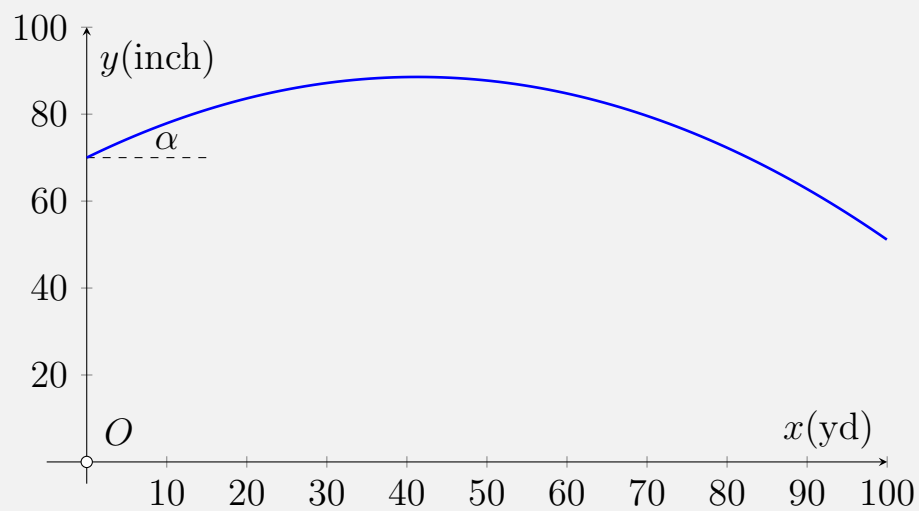
$$\begin{aligned} 0 &= \frac{70}{12} - \frac{51.2}{12} - \frac{9}{32}(32.2) + 300 \tan \alpha - \frac{9}{32}(32.2) \tan^2 \alpha \\ 0 &= -7.49 + 300 \tan \alpha - 9.06 \tan^2 \alpha \end{aligned}$$

The above expression is quadratic in $\tan \alpha$.

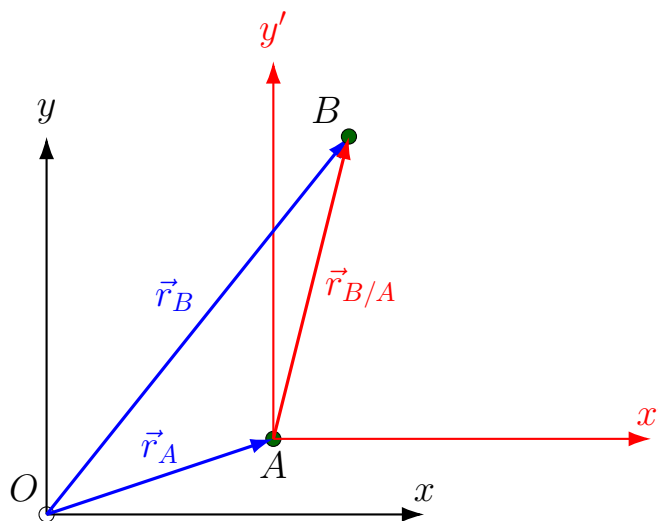
$$\begin{aligned} \tan \alpha &= 0.0250 & \tan \alpha &= 33.09 \\ \alpha &= 1.43^\circ & \alpha &= 88.3^\circ \end{aligned}$$

Both of these solutions are physical, but we are likely to choose the lower angle, which is what I've plotted below. In case you're curious, $t_{hit} = 0.75$ s.

3.2 – cont



4 Motion Relative to a Frame in Translation



Consider two particles, A and B . In our *fixed frame of reference*

O_{xy} particle A and B are located at $\vec{r}_A$ and $\vec{r}_B$, respectively.

Let us now consider another coordinate system whose origin coincides with the position of particle A . We call this coordinate system $A_{x'y'}$. We suppose that while the origin of the axes move with particle A their orientation remains unchanged (*i.e.* x' remains pointing right and y' always points up).

The position of particle B in the moving frame A is

$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A}$$

Taking time derivatives

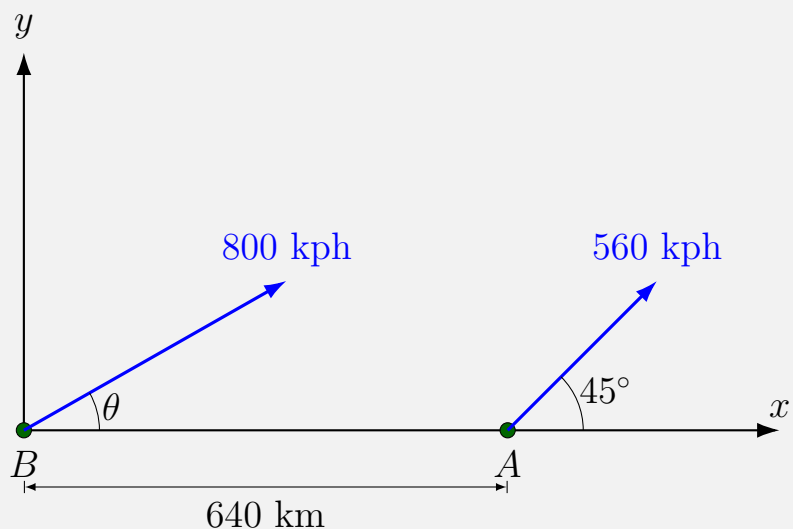
$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

Sample Problem 3.3

Airplane B is traveling at a constant velocity of 800 km/h, and is chasing airplane A, which is traveling northeast at a constant velocity of 560 km/hr. At time $t = 0$ s, airplane A is 640 km east of airplane B. Determine (a) the direction airplane B should travel (measured from the east) to intercept plane A; (b) how long it takes for airplane B to catch airplane A; (c) the rate at which the distance between the planes is decreasing.

3.3 – cont



Both airplanes have a constant velocity. These velocity are

$$\begin{aligned}\vec{v}_A &= (560 \cos 45) \hat{i} + (560 \sin 45) \hat{j} \\ \vec{v}_B &= (800 \cos \theta) \hat{i} + (800 \sin \theta) \hat{j}\end{aligned}$$

Integrating, the position is

$$\begin{aligned}\vec{r}_A &= [(560 \cos 45) t + 640] \hat{i} + [(560 \sin 45) t] \hat{j} \\ \vec{r}_B &= [(800 \cos \theta) t] \hat{i} + [(800 \sin \theta) t] \hat{j}\end{aligned}$$

Plane B intercepts plane A when $\vec{r}_A = \vec{r}_B$. We first set the y -components equal and solve for θ .

$$(800 \sin \theta) t = (560 \sin 45) t$$

3.3 – cont

The t cancels from both sides and we are left with

$$\sin \theta = \frac{(560 \sin 45)}{800} = 0.4950 \implies \theta = 29.67^\circ$$

Next we set the x -components equal and solve for t :

$$\begin{aligned}(560 \cos 45) t + 640 &= (800 \cos 29.67) t \\ 396t + 640 &= 695.1t \\ 640 &= (695.1 - 396) t = 299.1t \\ t &= \frac{640}{299.1} = 2.14 \text{ hr}\end{aligned}$$

The rate at which the distance is decreasing is the magnitude of $\vec{v}_{B/A}$. From above we have

$$\begin{aligned}\vec{v}_A &= 396\hat{i} + 396\hat{j} \\ \vec{v}_B &= 695.1\hat{i} + 396\hat{j}\end{aligned}$$

$$\begin{aligned}\vec{v}_{B/A} &= \vec{v}_B - \vec{v}_A = (695.1 - 396)\hat{i} + (396 - 396)\hat{j} = 299.1\hat{i} \\ |\vec{v}_{B/A}| &= 299.1 \text{ kph}\end{aligned}$$

Sample Problems

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2. An archer fires an arrow at a target that is 100 yards away. The target center is 51.2 inches off the ground. The arrow is released at a height of 70 inches from the ground with a speed of 400 fps. What firing angle α is required to hit the target?
 3. Airplane B is traveling at a constant velocity of 800 km/h, and is chasing airplane A, which is traveling northeast at a constant velocity of 560 km/hr. At time $t = 0$ s, airplane A is 640 km east of airplane B. Determine (a) the direction airplane B should travel (measured from the east) to intercept plane A; (b) how long it takes for airplane B to catch airplane A; (c) the rate at which the distance between the planes is decreasing.